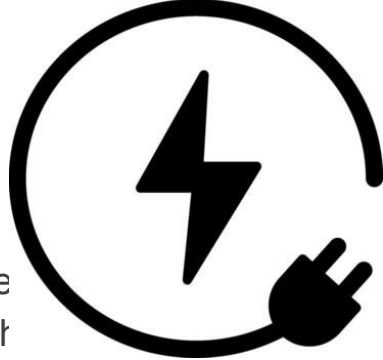
Week a Session A

Tuesday, Feb 24 2026

Overview

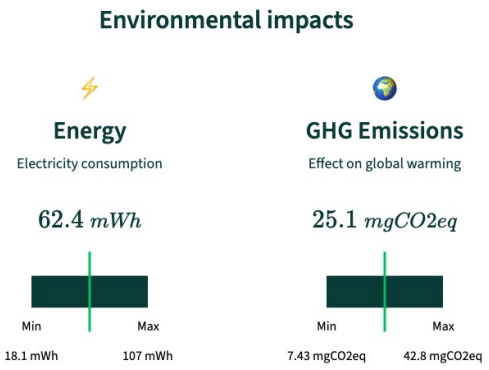
- Environmental Impact of AI: Energy
- Data Centers:
  - Data centers in our lives
  - Data centers in Lansing and in Saline Township
- Week 7/Spring Break Assignment

Environmental Impact of AI



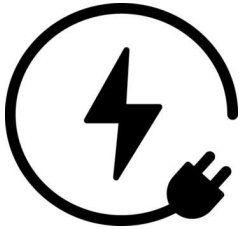
- **GHG Emissions:**
  - Gases that trap heat in the atmosphere are called greenhouse gases. The main type of gas is carbon dioxide (CO_2). It enters the atmosphere through burning fossil fuels, solid waste, etc.
  - The greenhouse gas allow sunlight to enter the atmosphere but prevent heat from escaping, which leads to global warming.
- **Where does GHG come from?**
  - Energy production (electricity), transportation, industry, etc.
- **Energy** used by data centers:
  - EcoLogits and its calculator

Environmental Impact of AI



- **Terms:**
  - mWh: milliwatt-hour, = 0.001 watt-hour. A smartphone battery is about 10,000–15,000 mWh (10–15 Wh). 62.4 mWh is 0.6% of 1% of a phone battery.
  - $mgCO_2eq$: megagrams of carbon dioxide equivalent. A Google search: 0.2–1 g CO_2 (200–1000 mg).

Environmental Impact of AI



- What are the goals of this?
- How to [reduce AI environmental impact](#)?

  - Look at **indirect impacts** of your project. Does the finality of your project is impacting negatively the environment?
  - **Be frugal** and question your usage or need of AI
    - Do you really need AI to solve your problem?
    - Do you really need GenAI to solve your problem? (you can read this paper)
    - Use small and specialized models to solve your problem.
    - Evaluate before, during and after the development of your project the environmental impacts with tools like 🌱 EcoLogits or CodeCarbon (see more tools)
    - Restrict the use case and limit the usage of your tool or feature to the desired purpose.
  - Do NOT buy new GPUs / hardware
    - Hardware manufacturing for data centers is around 50% of the impact.
  - Use cloud instances that are located in low emissions / high energy efficiency data centers (see electricitymaps.com)
  - Optimize your models for production
    - Quantize your models.
    - Use inference optimization tricks.
    - Prefer fine-tuning of small and existing models over generalist models.

Data Centers in Our Lives

- Activity:
  - We are adding a table to our presentation poster and on our future website. Complete the table below based on information found on this website: data centers map

Regions	Numbers	Notes
Virginia	570 in the whole state	128 in just the one town Ashburn
Michigan	70	7 in Lansing
New york	137	56 just in NYC
California	288	64 in Santa Clara

Local Data Centers

- **Class activity**:
  - Lansing: proposed by Deep Green
  - Saline Township: proposed by Open AI
  - Take notes of your findings in this document
- Societal response:
  - Economic Development Responsibility Alliance of Michigan (EDRA)
  - A documentary on Meta's data center

Week 7/Spring Break Assignment

- Data collection for our final project: Nexis Academic database